

RAI Risk Taxonomy 2.0:

Top-down evidence selection and staged classification of 1,711 global AI risks

RAI Risk Taxonomy Project

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Abstract

We present a reproducible, top-down account of the evidence and classification pipeline used to construct RAI Risk Taxonomy 2.0. The integrated evidence space contains 3,906,767 AI records: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. A curated AI-risk evidence view contains 82,971 records (2.1% of the AI evidence space). Canonicalization, source crosswalking, and risk-card construction yields 1,726 candidate rows. A conservative exact-match check merges one duplicate, yielding 1,725 candidate L4 cards. A taxonomy-level check against the 24 authoritative Physical AI L3 labels excludes 14 exact label leaks, yielding 1,711 L4 cards. A staged hierarchy-blind classification procedure initially assigns all cards to 50 L3 nodes. Subsequent definition and mapping audits, followed by a conservative v2.8.0 Agentic expansion, produce 54 L3 nodes and preserve an operational path for every card. The v2.12.0 release contains 1,165 General, 364 Agentic, and 182 locked Physical cards. Incremental constrained-EM audits preserve the authoritative Physical set and move only definition- and mechanism-supported boundary cases; 689 cards (40.3%) remain marked **HOLD** for human taxonomic review. BGE-M3 validation finds strong local support for the new families (93.5% top-1 containment) but does not establish global reliability of the inherited 54-family assignment (64.1% top-1 containment). We specify every evidence filter, classification rule, and validation limit.

Keywords: AI risk taxonomy; evidence synthesis; open-set classification; semantic mapping; Physical AI; responsible AI.

1 Introduction

A taxonomy can appear complete while concealing uncertainty in its source corpus, unit of analysis, or placement rules. RAI Risk Taxonomy 2.0 therefore separates four quantities that must not be conflated: the size of the integrated AI evidence space, the size of the AI-risk evidence view, the number of canonical L4 risk cards, and the number of L4 cards assigned to L3. The central result is a top-down evidence-to-taxonomy chain of 3,906,767 records, 82,971 AI-risk records, 1,726 candidate L4 rows, 1,725 exact-deduplicated candidates, 1,711 level-validated L4 cards, and 1,711 L3-assigned cards.

The transitions do not all preserve the same statistical unit. The first two stages count documents or patent records. The third and fourth stages count canonical risk concepts. Consequently, the 82,971 evidence records are not a one-to-one parent table from which every L4 card was mechanically generated. They form an auditable evidence view supporting corpus coverage and risk interpretation, whereas the L4 registry integrates risk taxonomies, benchmarks, standards, and linked evidence through a source crosswalk.

2 Top-down dataset construction

2.1 Stage A: integrated AI evidence space

The top-level dataset contains 3,906,767 records from three source families: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. Paper records originate from the local global Web of Science AI collections. Policy records combine the reviewed curated AI-policy master with the broader policy-materials collection. Patent records originate from the stage-2 AI patent master.

Records are standardized to a common schema and retained only when a non-empty title is available. Deduplication is performed within document type and analytical category. The ordered key hierarchy is

patent application ID for patents, followed by DOI, source record ID, URL, and a normalized title–year–institution fallback. Cross-category overlap is intentionally retained because AI, AI infrastructure, Physical AI, and AI risk are analytical views rather than mutually exclusive partitions.

2.2 Stage B: integrated AI-risk evidence view

The AI-risk evidence view contains 82,971 records: 50,375 papers, 938 policy reports, and 31,658 patents. Its inclusion rule is source-governed rather than a single keyword query. Papers combine curated academic risk references, the clean and integrated AI-risk paper corpora, Physical AI safety benchmarks, and verified paper records from the top-200 evidence set. Policy reports combine curated risk-policy references with verified report records. Patents combine curated risk-patent references with verified patent records. Record-type filters are applied before the same title-validity and within-category deduplication rules used at Stage A.

This view represents 2.1% of the integrated AI evidence space. Because the source collections and analytical categories can overlap, this percentage is a descriptive ratio, not the estimated prevalence of risk research in all AI scholarship.

2.3 Stage C: canonical L4 registry

The unit changes from evidence record to risk concept. Source risk IDs are joined through the source crosswalk, semantically redundant cards are reviewed, and one canonical row is retained per L4 failure mode or harm mechanism. Permanent identifiers RAI4–0001 through RAI4–1726 are allocated independently of the risk label. The initial result is 1,726 candidate L4 rows. Automatic deduplication is restricted to pairs whose normalized label and mechanism-only definition are both exactly equal; semantic-similarity thresholds are not used. This merges RAI4–1606 into RAI4–1083, retains both source references, and yields 1,725 exact-deduplicated candidates. The retired ID remains in the crosswalk. The registry includes the 182-card Physical AI taxonomy as a protected mapping set.

The ratio of L4 cards to AI-risk evidence records is 2.1%. It is reported only as a scale-change diagnostic: one document may support multiple risks, and one risk may be supported by multiple documents. It is not a document acceptance rate.

2.4 Stage D: taxonomy-level validation

The deployed Physical AI site is the authoritative source for its 24 L3 names and 182 L4 cards. Its live HTML was byte-identical to the local repository snapshot used for this check. We compare the English component of each authoritative Physical L3 name with every candidate L4 label after NFKC normalization, case folding, ampersand normalization, and removal of non-alphanumeric separators. A card is excluded only when these normalized labels are exactly equal; semantic similarity and definition similarity are not used. Fourteen L4 rows match category names and are retired as taxonomy-level leakage, including RAI4–0553 “Accidental harm”, which maps to Physical L3 P3.1. Their definitions and references remain in the audit crosswalk. This yields 1,711 canonical L4 cards without changing any of the protected 182 Physical L4 mappings.

2.5 Authoritative Physical L4 content synchronization

Release v2.4 synchronizes the content of all 182 protected Physical cards from the local repository underlying the deployed Physical AI Risk Taxonomy site. The immutable RAI4 crosswalk and L3 paths are retained, while the bilingual label and definition, severity, probability, 3H1R attributes, evidence justifications, and 360 reference rows are replaced by the current Physical source values. Of these reference rows, 359 contain a justification and one source row retains an empty justification without imputation. Bilingual terminal parenthetical text is split into Korean and English fields to preserve the global card schema. Impact is recalculated as $I = sp$ from the synchronized severity s and probability p . Percentile is not displayed for any card because the cards do not constitute a common measurement population suitable for relative ranking. Existing percentile fields remain provenance data only.

2.6 Complete English–Korean localization

Release v2.5 requires bilingual display at every taxonomy level. The supplied L0–L3 terminology and Korean definitions are used as the controlled glossary. The 182 Physical L4 cards retain their authoritative bilingual source text. For the remaining 1,529 cards, the localization pipeline removes the repeated provenance boilerplate from each English definition, selects the first sentence that states the core failure mechanism, generates a concise Korean definition, and normalizes risk-domain terminology. Context-sensitive corrections include *poisoning* as “data or model contamination” rather than substance addiction, *washing* as false compliance presentation, and *rubber stamping* as nominal approval. Every generated row retains the English source-sentence hash, method, and `human_review_status=pending`. Thus all 1,711 cards contain non-empty English and Korean labels and definitions, while automated Korean drafts remain explicitly reviewable rather than being represented as human-approved ground truth.

2.7 Canonical L2 consolidation

Release v2.6 normalizes L2 to three canonical categories: *Interaction Safety*, *System Safety*, and *Societal Safety*. The former General labels *Interaction* and *System*, and the former Agentic label *System*, are renamed accordingly. Six L1-specific path nodes remain as implementation edges so that the strict L0–L1–L2–L3 tree and all L3 parent paths are preserved; each edge references one of the three canonical L2 identifiers. Thus the reported L2 category count is three, while no L3 path or L4 assignment changes.

2.8 Stable L4 identifiers and L1 display semantics

The public explorer keeps the stable `RAI4-####` identifier independent of taxonomy placement. L1 membership is communicated through an explicit text badge, a consistent icon, and a visual color: a brain and blue for General-purpose AI, a compass and green for Agentic AI, and a robot and red for Physical AI. The icon–color pair is repeated in the global navigation, filters, hierarchy panel, and L4 cards. The color is derived from the card’s current L1 path at render time rather than encoded into the permanent identifier. This allows later human-approved remapping without identifier churn and avoids using color or icon as the sole carrier of categorical information.

2.9 Stage E: L3 assignment policy and later audits

The operational policy assigns all 1,711 canonical L4 cards to an L3 node (100.0%). The original three-stage classifier used 50 L3 nodes and retained 76 decision-review markers after level validation. Release v2.7.0 then removed 1,129 instances of circular taxonomy prose from non-Physical definitions, performed evidence-constrained mapping review, and marked 692 cards where the available definition and evidence did not justify an approved taxonomic fit. The v2.8.0 Agentic expansion evaluates every non-Physical card while locking all 182 Physical mappings. It moves 31 cards into four new L3 nodes and clears seven review markers, leaving 685 cards (40.0%) marked **HOLD**. Releases v2.9.0–v2.12.0 then audit Anthropomorphism, Goal Misalignment, and Copyrights, leaving 689 cards (40.3%) marked **HOLD**. The marker is internal review metadata on an already assigned card, not a separate L1, L2, or L3 category.

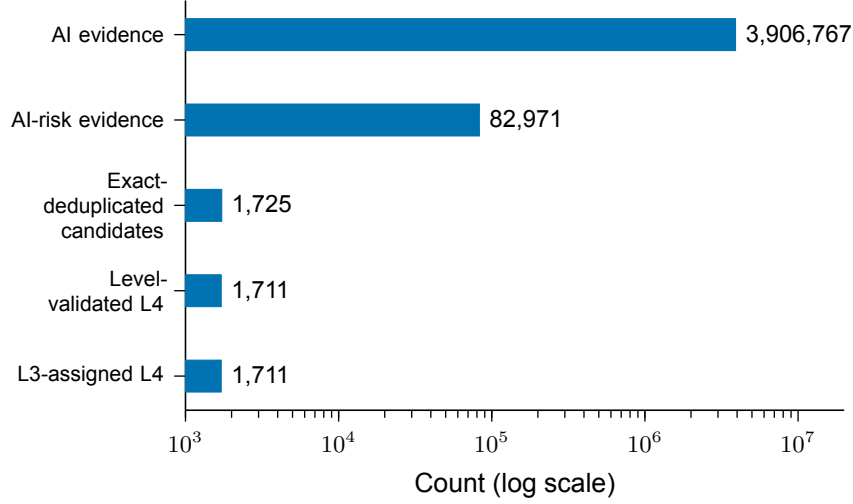


Figure 1: **Top-down evidence and taxonomy funnel.** Counts move from document/patent records to canonical risk concepts at the transition from AI-risk evidence to L4. The logarithmic axis preserves visibility across four orders of magnitude.

Table 1: Top-down stages, counts, and inclusion filters.

Stage	Count	Unit	Principal filter
AI evidence	3,906,767	Document or patent record	AI source collections; valid title; standardized schema; within-type/category deduplication.
AI-risk evidence	82,971	Document or patent record	Curated/integrated risk sources; record-type inclusion; valid title; within-type/category deduplication.
Exact-deduplicated candidates	1,725	Risk card	Exact normalized label and core-definition match only; one retired ID mapped to its canonical card.
Canonical L4	1,711	Risk card	Exact normalized-label exclusion against 24 authoritative Physical L3 names; 14 level leaks retired; all 182 Physical L4 cards preserved.
L3 assigned	1,711	Risk card	Protected Physical mapping or staged non-Physical placement; 689 decision-required cards remain assigned and marked HOLD.

3 Mathematical classification procedure

3.1 Protected Physical AI mapping

The 182 L4-to-L3 mappings inherited from the Physical AI taxonomy are treated as the locked gold set. Later stages cannot change these paths. Before the taxonomy-level exclusion, the remaining 1,543 candidates enter the hierarchy-blind classification procedure. Legacy global L0–L3 labels, source families, and source-ID prefixes remain reference metadata and are excluded from prediction features.

3.2 Stage 1: strict open-set gate

The input for each non-Physical card is the L4 label, a mechanism-only definition with hierarchy restatements removed, and the evidence title. Let s_{sem} be the top-1 semantic score, s_{comp} the top-1 composite score, m_{sem} and m_{comp} their respective top-1 margins, and v_{anchor} the number of anchor views selecting the same L3. A candidate is retained only when all seven conditions hold:

1. exactly one L3 is rule-eligible;

2. the rule winner, semantic top-1, and composite top-1 are identical;
3. $s_{sem} \geq 0.55$ and $s_{comp} \geq 0.54$;
4. $m_{sem} \geq 0.035$ and $m_{comp} \geq 0.035$;
5. $v_{anchor} \geq 2$ across three anchor views;
6. no gap sentinel, multi-mechanism flag, or Physical-boundary flag is present; and
7. two independent expert agents approve the same exact L3.

This gate yields 37 new expert-consensus proposals. Together with the 182 Physical locks, Stage 1 assigns 219 cards and withholds 1,507.

3.3 Stage 2: suitability-ranked relaxation

Stage 2 preserves all 219 protected assignments and computes

$$S_2 = 0.55q_{sem} + 0.20q_{sm} + 0.10q_{cm} + 0.10q_{vote} + 0.05q_{rule} - p_{gap}, \quad (1)$$

where

$$q_{sem} = \text{clip} \left(\frac{s_{sem} - 0.45}{0.25}, 0, 1 \right), \quad (2)$$

$$q_{sm} = \text{clip} \left(\frac{m_{sem}}{0.05}, 0, 1 \right), \quad (3)$$

$$q_{cm} = \text{clip} \left(\frac{m_{comp}}{0.05}, 0, 1 \right), \quad (4)$$

$$q_{vote} = v_{anchor}/3, \quad (5)$$

$$q_{rule} = \mathbf{1}[\text{at least one eligible L3}], \quad (6)$$

$$p_{gap} = 0.05 \mathbf{1}[\text{gap sentinel present}]. \quad (7)$$

S_2 is a deterministic ranking score rather than a calibrated probability. Stage 2 first preserves 55 hard holds. It then orders the remaining holds by ascending S_2 , with permanent RAI4 ID as the tie breaker, and reserves the lowest 118. Thus $55 + 118 = 173$ cards remain unresolved and 1,553 cards are assigned (90.0%).

3.4 Stage 3: forced-path test and final policy

Stage 3 preserves the 1,553 Stage 2 paths and assigns only the 173 unresolved cards to the hierarchy-blind top-1 L3 already stored in the score artifact. This produces 1,726 operational paths but does not validate taxonomy fit. Each new path retains its original hold reason, score, review priority, and `human_approved=false` marker.

The final policy marks five difficult-case families for decision review: 23 Physical-scope reviews, 17 expert rejections, 7 expert disagreements, 6 multi-mechanism cases, and 2 low-absolute-fit cases. Their forced L3 path remains populated and `decision_required=true`. The remaining 118 quota holds also retain their forced paths and review metadata.

3.5 Independent expert card review

Two reviewers independently inspected all 1,725 cards for agreement among the L4 label, mechanism definition, evidence, and assigned L3. Reviewer A proposed 30 amendments and Reviewer B proposed 14. Automatic amendment required both reviewers to propose the same label and L3 at high confidence. One card met this rule: RAI4-0888 changed from “Privacy concerns” to “Anthropomorphic trust-induced privacy disclosure”, while retaining RAI3-G-INT-10. Its original label remains in provenance metadata. Forty-two cards had a proposal from only one reviewer or incompatible proposals; their labels and paths were not

changed. Thirteen were already marked **HOLD** and 29 newly received the marker. The consensus amendment cleared the prior hold on **RAI4-0888**, giving $55 - 1 + 29 = 83$ final **HOLD** cards. No protected Physical AI card was changed.

The later taxonomy-level validation supersedes the proposed adjudication of **RAI4-0553**: “Accidental harm” is itself the authoritative Physical L3 category **P3.1**, not an L4 failure mode. It is therefore excluded rather than renamed or remapped. Thirteen additional exact Physical-L3 label leaks are handled identically. Six of the excluded rows previously carried **HOLD**, reducing the displayed count from 82 to 76. The protected Physical set remains exactly 182 cards.

3.6 v2.7 definition audit and v2.8 Agentic expansion

The v2.7 audit treats L3 load as a triage signal rather than evidence of semantic fit. It removes 1,129 imported hierarchy-name scaffolds from L4 definitions, applies nine evidence-constrained remaps, and marks 692 cards for decision review. Release v2.8.0 then adds four Agentic-specific L3 families: *Goal & Planning*, *Tool Calling*, *Memory*, and *Oversight & Control*. Each node has an English–Korean definition and explicit references. A non-Physical card moves only when its mechanism is intrinsically Agentic, the new family is more specific than every existing General or Agentic alternative, and the necessity gate rejects adequate coverage by the prior taxonomy. Formally, for card i and new family k ,

$$k_i^* = \arg \max_{k \in K_{new}} s(i, k), \quad \Delta_i = s(i, k_i^*) - \max_{j \in K_{old}} s(i, j), \quad (8)$$

and movement additionally requires mechanism-specific rules, sufficient absolute fit, positive separation, and no Physical lock. The deterministic iteration changes 31 paths on pass 1 and zero on pass 2. Counts in the new families are 11, 11, 3, and 6, respectively. All other placements are preserved, and the v2.8.0 review queue contains 685 cards.

3.7 Incremental constrained-EM audits, v2.9.0–v2.12.0

Later audits do not apply unconstrained nearest-centroid output. Each audit allows only the selected source L3 cards to move, treats all other cards as fixed anchors, excludes every Physical destination, and recomputes candidate centroids until no assignment changes. The composite score is

$$S(i, k) = 0.60 c(i, k) + 0.30 d(i, k) + 0.10 w(i, k), \quad (9)$$

where c is current-centroid cosine similarity, d is L3-definition cosine similarity, and w is TF–IDF keyword cosine similarity. A destination must improve on the source by at least 0.020 and have keyword cosine of at least 0.015. Agentic destinations additionally require an explicit autonomous execution mechanism, and destination-specific definition guards reject matches caused by generic shared terms.

The cumulative audits move three Anthropomorphism cards in v2.9.0, five Goal Misalignment cards in v2.10.0, four Copyrights cards in v2.11.0, and eight additional Anthropomorphism cards in v2.12.0. Every such move remains **HOLD** until human adjudication. The final domain counts are 1,165 General, 364 Agentic, and 182 Physical cards; the review queue contains 689 cards.

4 HOLD decision policy

4.1 Meaning and state transition

HOLD is the public rendering of `decision_required=true`. It is not an unclassified bucket and does not create another L1, L2, or L3. A held card always retains exactly one operational L3 path, but that path must not be treated as human-approved ground truth. The state is set by a hard qualitative trigger or by a pending algorithmic-review trigger. It is cleared only by a recorded human decision that reconciles the label, mechanism-only definition, evidence, and destination definition; absence of a competing score alone is not sufficient.

4.2 Quantitative criteria

Quantitative measurements control automatic eligibility and review priority, not semantic truth. The following rules are applied or reported:

1. Stage 1 automatic acceptance requires $s_{sem} \geq 0.55$, $s_{comp} \geq 0.54$, both margins at least 0.035, at least two of three anchor votes, exact agreement of the rule, semantic, and composite winners, and two matching expert-agent approvals. Failure prevents automatic clearance.
2. Stage 2 uses the deterministic suitability score S_2 defined above; the lowest-ranked reserve and every hard hold retain decision metadata after forced operational assignment.
3. A constrained-EM move requires composite improvement of at least 0.020 over the source and keyword cosine of at least 0.015. Passing these thresholds permits a proposal but never clears HOLD automatically.
4. For v2.12.0 audit prioritization, a top-two cosine margin below 0.020 is high priority, a margin from 0.020 to below 0.050 is medium priority, and a margin of at least 0.050 is lower priority. Released-L3 rank outside the top three or per-card agreement below 80.0% under $\sigma = 0.05$ perturbation is an additional instability signal.
5. Strong geometric support is reported only when top-1 containment is at least 90.0%, top-2 containment at least 95.0%, the matched-size permutation test has $p < 0.001$, and perturbation agreement remains at least 97.0% through $\sigma = 0.05$. These are validation thresholds, not substitutes for the qualitative tests below.

4.3 Qualitative criteria

A card remains or becomes HOLD when any of the following substantive conditions applies:

- the label and mechanism definition describe different risks, or the cited source does not directly support the stated failure mode;
- the card is multi-mechanism, the destination L3 is overloaded, or two plausible L3 paths cannot be separated without a policy judgment;
- the risk exposes a taxonomy gap, so the present L3 is merely the closest operational path rather than a definitionally adequate family;
- an Agentic destination lacks an intrinsically Agentic execution loop, tool action, persistent memory, autonomous planning, or oversight mechanism;
- a non-authoritative card appears Physical but is outside the locked 182-card Physical set, or a proposed change would disturb a locked Physical path;
- independent reviewers reject the proposal, disagree on the destination, or do not reach exact high-confidence consensus; or
- the card was moved by constrained EM and has not yet received explicit human approval.

4.4 Current v2.12.0 HOLD composition

The 689 held cards are grouped by the stored primary reason in Table 2. Composite taxonomy-gap labels are counted once in the taxonomy-gap row.

Table 2: Primary reasons for the v2.12.0 HOLD queue.

Primary reason group	Cards
Taxonomy gap	280
Anthropomorphism mechanism not established	173
Overloaded L3 or low evidence fit	138
Expert review without consensus	28
Constrained-EM remap pending review	20
Physical-boundary conflict	16
Expert rejection	16
Expert disagreement	7
Multiple mechanisms	5
Other named exceptions	4
Low absolute fit	2
Total	689

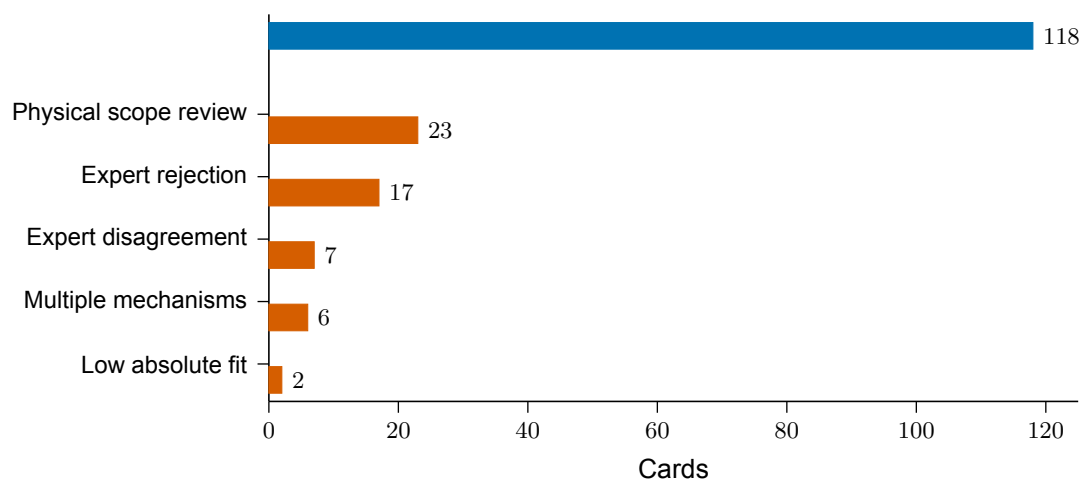


Figure 2: **Disposition of the 173 Stage 2 holds.** The 55 difficult cases (vermillion) are force-assigned and marked HOLD; 118 suitability-reserve cards (blue) are also force-assigned.

5 Results

5.1 Evidence composition

Papers account for 44.8% of the integrated AI evidence space, policy reports for 3.2%, and patents for 52.0%. Within the AI-risk evidence view, the corresponding shares are 60.7%, 1.1%, and 38.2%. These shares use one decimal because their integer component is at least one.

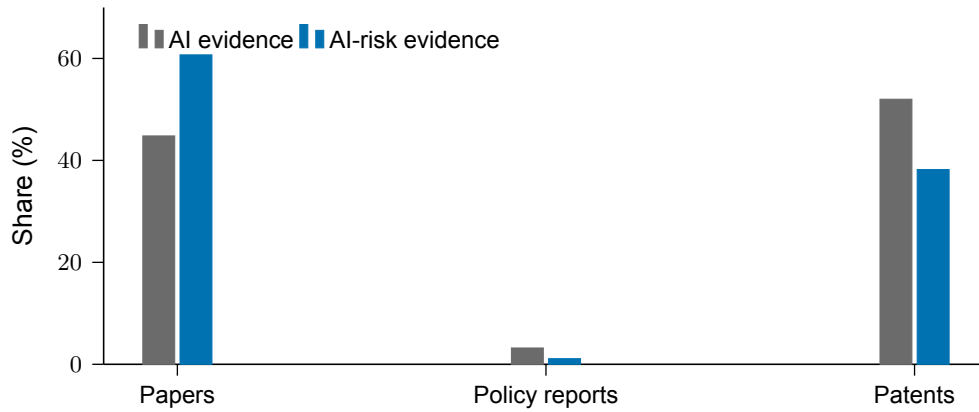


Figure 3: **Source-family composition.** Percentages compare document-type composition within the integrated AI evidence space and the AI-risk evidence view.

5.2 Final hierarchy coverage

The final policy assigns all 1,711 canonical cards (100.0%); 689 assigned cards (40.3%) carry the **HOLD** marker. Of all cards, 1,165 are General, 364 are Agentic AI, and 182 are Physical AI. Their shares are 68.1%, 21.3%, and 10.6%, respectively. All 54 L3 nodes contain at least one card. Anthropomorphism is the largest L3 with 234 cards (13.7%); the five largest L3 nodes jointly contain 40.4%.

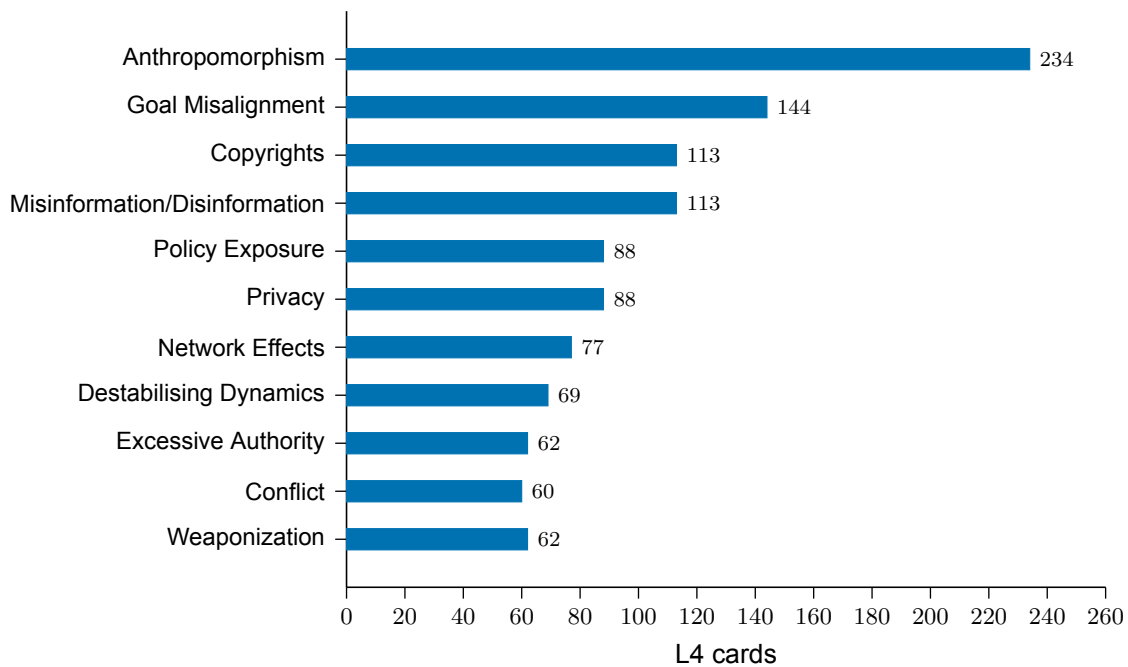


Figure 4: **Largest L3 nodes in the final policy view.** The figure shows the eleven largest L3 nodes. Counts include assigned cards marked **HOLD**.

6 Rounding and reporting policy

All counts are reported as integers. Percentages below 1 are rounded at the third decimal place and displayed to two decimal places; percentages with an integer component of at least 1 are rounded and displayed to one decimal place. For example, the L4 registry represents 0.04% of the integrated AI evidence count, the AI-risk view represents 2.1%, final L3 coverage is 100.0%, and the decision-required share is 40.3%. Exact floating-point ratios remain available in the machine-readable statistics artifact.

7 Assignment reliability validation

7.1 Reproducible procedure

Reliability is evaluated on all 1,711 published paths using dense, 1,024-dimensional BGE-M3 embeddings, seed 20260704, maximum sequence length 256, batch size 32, and L2 normalization. Card text concatenates bilingual L4 labels and definitions; L3 seed text concatenates bilingual L3 labels and definitions. The procedure reproduces four checks from the prior Technical Report.

First, seed-initialized spherical EM alternates nearest-centroid assignment and normalized centroid re-computation:

$$z_i^{(t+1)} = \arg \max_k y_i^\top \mu_k^{(t)}, \quad \mu_k^{(t+1)} = \frac{\sum_{i:z_i=k} y_i}{\|\sum_{i:z_i=k} y_i\|_2}. \quad (10)$$

Second, top- k containment is the proportion of cards whose published L3 is among the k nearest centroids calculated from the published partition. Third, the observed mean within-family cosine is compared with 5,000 label permutations that exactly preserve family sizes; the p-value uses the plus-one correction. Fourth, isotropic Gaussian embedding perturbation is repeated 200 times per σ , and agreement compares perturbed with unperturbed nearest-centroid assignments.

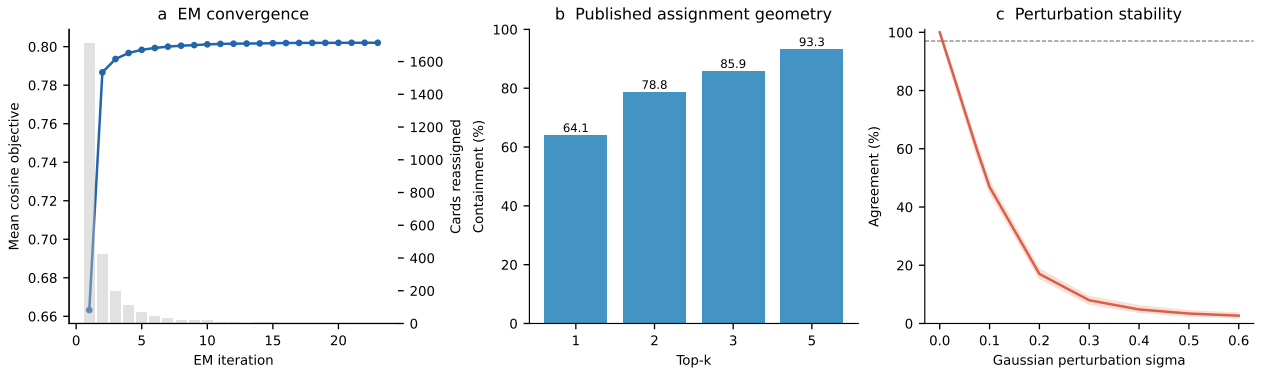


Figure 5: **Convergence and stability of the v2.8.0 assignment.** (a) Seed-initialized spherical EM reaches a fixed point after 23 iterations. (b) Published-family top- k containment remains below the strong global criteria. (c) Agreement declines as Gaussian embedding noise increases.

Table 3: Reliability results and v2.7.0 comparator.

Metric	v2.7.0	v2.8.0	New-family cards
L3 families	50	54	4
Cards assessed	1,711	1,711	31
Mean within-family cosine	0.768	0.770	0.837
Top-1 containment	64.4%	64.1%	93.5%
Top-2 containment	79.0%	78.8%	96.8%
Top-3 containment	86.4%	85.9%	96.8%
Top-5 containment	93.3%	93.3%	—
Permutation p-value	< 0.001	< 0.001	—
Agreement at $\sigma = 0.01$	94.6%	94.6%	—
Agreement at $\sigma = 0.05$	72.7%	73.0%	—

The EM objective increases monotonically from 0.663 to 0.802 and reaches zero reassignments after 23 iterations. The converged unsupervised partition, however, agrees with only 20.2% of published paths. Numerical convergence is therefore not evidence that the published semantics are correct. Published cohesion is 0.770, above the matched-size null mean of 0.739 with zero exceedances in 5,000 permutations (plus-one $p = 0.0002$). The taxonomy contains non-random semantic structure, but its global partition is not robust under the tested geometry.

7.2 Four new Agentic families

Table 4: Geometric support for the four new Agentic L3 families.

L3 family	Cards	Mean cosine	Top-1	Top-3
Goal & Planning	11	0.805	81.8%	90.9%
Tool Calling	11	0.839	100.0%	100.0%
Memory	3	0.946	100.0%	100.0%
Oversight & Control	6	0.839	100.0%	100.0%

Across the 31 moved cards, top-1 containment is 93.5%, top-2 and top-3 are 96.8%, and the median margin over the strongest competing centroid is 0.093. Tool Calling, Memory, and Oversight & Control are geometrically unambiguous in this release. Goal & Planning contains two top-1 exceptions, including one outside top-3, and receives priority human review.

7.3 Decision and limitations

New Agentic expansion: PASS WITH REVIEW. Three new families achieve 100.0% top-1 containment and the moved subset is substantially more coherent than the global mean. The Goal & Planning exceptions remain review items.

Full 54-family taxonomy: NOT YET DEMONSTRATED. The permutation test establishes non-random structure, but global containment, perturbation stability, and EM-to-published agreement do not satisfy strong reliability criteria. The weakest top-1 families are Destabilising Dynamics (36.2%), Goal Misalignment (38.9%), Network Effects (40.3%), Copyrights (50.4%), Conflict (53.3%), and Excessive Authority (56.5%). Anthropomorphism remains large and heterogeneous at 245 cards and 63.3% top-1 containment.

Four limitations constrain interpretation. First, cross-category overlap in the evidence space means that source counts are analytical views, not mutually exclusive samples. Second, the transition from 82,971 records to 1,711 L4 cards changes the unit of analysis. Third, embedding geometry is a diagnostic, not a substitute for expert judgment or causal evidence. Fourth, small families can obtain perfect containment with limited statistical power.

7.4 v2.12.0 non-Physical Algorithm 2 validation

The executable notebook applies the Technical Report’s seed-initialized spherical EM procedure to all 1,529 non-Physical cards and the 30 non-Physical L3 families. It excludes the 182 locked Physical cards and does not mutate the release. The objective increases monotonically and reaches a fixed point after 18 iterations with no empty L3 family. Numerical convergence again does not establish taxonomic validity: exact agreement with the released paths is 30.8%, adjusted Rand index is 0.149, normalized mutual information is 0.351, and unconstrained EM proposes 1,058 changes (69.2%).

The current partition remains significantly more cohesive than matched-size random partitions: observed mean cohesion is 0.765 versus null mean 0.738, with zero exceedances in 5,000 permutations and plus-one $p = 0.0002$. Yet perturbation agreement for the EM partition is only 85.2% at $\sigma = 0.05$, below the 97.0% strong-stability criterion. Accordingly, the 1,058 changes are exported only as a diagnostic review queue. They are not applied to v2.12.0.

8 Review priorities

Review should begin with the geometric exceptions in Goal & Planning, then the six weakest global families and low-margin cards within Anthropomorphism. The 689 assigned cards marked **HOLD** should be adjudicated using agreement among the risk label, mechanism-only definition, cited evidence, and candidate L3 definition. Each approved batch should be followed by the identical reliability analysis. A low score alone

must not trigger automatic movement; human review determines whether to retain the path, expand an L3 definition, or propose a genuinely necessary new L3.

Data availability

The exact top-down counts, filters, placements, and public card bundle are available in the repository:

- `reports/statistics/top_down_evidence_funnel.json`
- `data/experiments/stage3-forced-policy-v2/placements.json`
- `public/data/releases/v2.12.0/cards.json`
- `public/data/releases/v2.12.0/hierarchy.json`
- `reports/data_quality/l4_deduplication_v2.1/retired_to_canonical.json`
- `reports/expert_review/v2.1/consensus_result.json`
- `reports/data_quality/physical_l3_label_leakage_v2.3.json`
- `reports/data_quality/physical_card_sync_v2.4.json`
- `reports/localization/nonphysical_l4_ko_v2.5.json`
- `reports/data_quality/l2_category_consolidation_v2.6.json`
- `reports/data_quality/l4_definition_mapping_remediation_v2.7.0/summary.json`
- `reports/data_quality/agentic_l3_expansion_v2.8.0/summary.json`
- `reports/validation/v2.8.0/reliability/reliability_summary.json`
- `reports/validation/v2.8.0/reliability/per_l3_metrics.json`
- `reports/validation/v2.12.0/em_all_nonphysical/reliability_summary.json`
- `reports/validation/v2.12.0/em_all_nonphysical/candidate_remaps.csv`
- `output/jupyter-notebook/algorithm2_em_l3_nonphysical_validation_v2_12.ipynb`

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